<u>Element A</u>	<u>Information</u> **Element A: Presentation and Justification of the Problem**
Problem Statement and Details (A1, A2, A5) **Research**	***Explain your problem in detail, including details that you might consider common knowledge. Use data and statistics to justify your problem statement. Do not provide a bulleted list of facts; rather, organize the problem details in paragraph form. Address as many of the 5 Ws +1 H (who, what, where, when, why, how) as is possible, for example:*** ***1) Who is involved in this problem?*** ***2) What are they doing?*** ***3) What unique situations does the problem occur within, if any?*** ***4) What is the size of the issue?*** ***5) Where is the problem happening?*** ***6) When does the problem occur?*** ***7) Why does the problem happen?*** ***8) How often does the problem happen?*** We are planning a project to solve a current problem occurring in our art wing. The art students and art teachers have noticed a problem with fumes and contaminants in the air that distracts them from work and can also be dangerous. This current situation is unique because of the limited space available for our filter. Our stakeholders, Mrs. Poole and Mrs. Hudson, have requested the filters be mounted on the wall or ceiling. They have also requested that the design be sleek and aesthetically pleasing, so as not to disturb the current layout or design of the room. The problem occurs when the art students are having class and soldering, 3D printing, woodburning, using resin, or grinding glass. Though Mrs. Poole uses a low VOC resin, the dangers of breathing in resin fumes are still present. These dangers include "irritation of the eyes, nose, throat, and skin, skin allergies, and asthma"(IP Systems USA). The problem of air purification is very relevant, especially concerning art studios. The benefits of air purification in an art studio include reduced respiratory issues and getting rid of allergens, dust, bacteria, and viruses. This problem occurs in our art room 2 to 3 times a week and the fumes can last multiple hours. https://ipsystemsusa.com/hazards-of-inhaling-epoxy-fumes/ Air quality becomes a problem in cases in which pollutant particles are excessively present in an area. According to the Center for Disease Control (CDC), particulate matter is made up of "particles of solids or liquids in the air" including "dust, dirt, soot, smoke, and drops of liquid" (National Center for Environmental Health). In the case of the art wing, dust from glass cutting and ceramics, smoke from woodworking, and various scents from resin and melted plastics. Air pollution, in our case of the

	art wing and as a whole, is a **major** issue, especially as it affects the health of those within the polluted environment. While anyone can be affected by particulate matter, “people with heart or lung diseases” and “older adults” can experience symptoms beyond the imminent effects, including “eye irritation, lung and throat irritation” as well as long-term consequences including “lung cancer” (National Center for Environmental Health). Considering the frequent presence of dust and smoke in twenty classes per week, poor air quality in the art wing must be solved as soon as possible to preserve the health and safety of students and teachers alike. The problem occurs when people use resin or when they 3D print. This is because when they happen, in resin they emit VOC, and the plastics emit ABS and PLA. These are harmful because they can negatively affect our health. Volatile organic compounds (VOCs) can be emitted as gasses from things like 3D printers (United States Environmental Protection Agency). Negative health effects can include, “Eye, nose and throat irritation, headaches, loss of coordination and nausea, damage to liver, kidney and central nervous system,” and possibly “cancer,” also, “key signs or symptoms associated with exposure to VOCs include: conjunctival irritation, nose and throat discomfort, headache, allergic skin reaction, dyspnea, declines in serum cholinesterase levels, nausea, emesis, epistaxis, fatigue,” and “dizziness,” (United States Environmental Protection Agency). PLA can have negative effects, because even though it is organic, according to Gianeco, a website that specializes in sustainable printing products, “it has higher permeability than other plastics”(Gianeco). When 3D printers heat plastics, such as ABS plastics, they are released into the surrounding environment as fumes. According to ALL3DP, a website specializing in all things 3D printing, common symptoms of this are “drowsiness, eye irritation, nausea, and headaches.” “Not only are fumes from ABS more hazardous than PLA, but ABS also releases more of these harmful fumes while printing. This is attributed to the higher printing temperatures demanded by ABS printing, which is typically between 220-260 °C, as opposed to PLA, which prints between 180-220 °C,” (ALL3DP). This is a daily problem for Mrs. Poole, Mrs. Hudson, and their students. The clay dust from ceramics, the broken down glass in glass, as well as the resin, PLA, and ABS materials from 3d printing cause the most issues within the class. A lack of air circulation, as well as none of the rooms being able to open their windows makes this a constant problem for every student. Kyle ***Resin, (specifically Eproxy, formed by evaporated resin) is severely toxic when inhaled. It can lead to difficulty or rapid breathing, drooling, eye pain, pain in the mouth or throat, or***

	hoarseness in the mouth or throat. People with asthma are at most risk of being effected ***In order to earn a score of 5, you will need to cite at least five sources that are from the last ten years and that come from 3 or more of the following credible categories: peer reviewed journals, professional organizations, government agencies, local and national newspapers. Use in-line citations to explicitly state which information is from which source. Do not just list all of the citations at the end of the submission or in footnotes.***
Stakeholder Concerns and Connections (A3, A4) **Primary Stakeholders and concerns** **Secondary Stakeholders and concerns**	***Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem. Consider the breadth of the problem and how it might have social and environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system.*** The two primary stakeholders in our project are Mrs. Poole, Mrs. Hudson. The concerns of our primary stakeholders regard space and design. Mrs. Poole would like the design to be visually appealing while also doing its job. Both teachers want a design that can maximize the space of their rooms and be out of the way while students are working. Secondary stakeholders include: Mrs. Boone, art students, students in neighboring classrooms, Mr. Buffone, and Dr. Jackson. The secondary stakeholders, the students, are also being impacted by our air filter. They are also being exposed to the fumes and will benefit immensely from a functioning air filter.
Problem Importance (A6)	***End Element A with a compelling argument for why this problem is important and needs to be solved now. Support this argument with data and statistics.*** Fumes and particles in the art wing can have both long-term and imminent consequences on the health of students and faculty as well as the overall classroom environment, ranging from "lung and throat irritation" to "lung cancer" that can be particularly dangerous to the older adults and individuals with heart/breathing problems in the Catholic High community (National Center for Environmental Health). Our goal is to provide a filter that produces clean air. Clean air is "air that has no harmful levels of pollutants (dirt and chemicals) in it" (AirNow.gov). Solving this problem is crucial

	to the health and well being of art students and teachers, but also all students and teachers at Catholic High School. Through solving this problem and making a filter, we hope to improve the safety and happiness of all students and teachers. https://www.airnow.gov/education/students/clean-and-dirty-air-part-one/#:~:text=Clean%20air%20is%20air%20that,and%20have%20a%20bad%20smell.

Member Key:

- Ryan
- Sophia
- Kyle
- Sam

URLs

(United States Environmental Protection Agency):
https://www.epa.gov/indoor-air-quality-iaq/volatile-organic-compounds-impact-indoor-air-quality#:~:text=Volatile%20organic%20compounds%20(VOCs)%20are,ten%20times%20higher

(National Center for Environmental Health):
https://www.cdc.gov/air/particulate_matter.html#:~:text=Particle%20Pollution%20and%20Your%20Health&text=Dust%20from%20roads%2C%20farms%2C%20dry,or%20even%20into%20your%20blood.

(AirNow.gov):
https://www.airnow.gov/education/students/clean-and-dirty-air-part-one/#:~:text=Clean%20air%20is%20air%20that,an d%20have%20a%20bad%20smell.

(IP Systems USA):
https://ipsystemsusa.com/hazards-of-inhaling-epoxy-fumes/

https://www.mountsinai.org/health-library/poison/plastic-casting-resin-poisoning

(Gianeco):
https://www.gianeco.com/en/faq-detail/1/6/what-are-the-disadvantages-of-pla#:~:text=Because%20PLA%20is%20made%20of,term%20storage%20of%20food%20products.

(ALL3DP)
https://all3dp.com/2/is-abs-plastic-safe-toxic/